

A Novel Comprehensive Water Hammer Pressure Model for Fracture Geometry Evaluation

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Summary

Hydraulic fracturing is a critical technology in reservoir modification design. Post-fracturing fracture geometry characterization is of essential importance for the design and optimization of hydraulic fracturing treatments. In recent years, fracture geometry evaluation methods based on water hammer pressure analysis have gained significant attention due to their cost-effectiveness and operational simplicity. However, existing water hammer models based on equivalent circuit theory exhibit limited physical interpretability between model parameters and fracture geometry characteristics. In this study, we propose a novel water hammer modeling approach with enhanced physical interpretability of model parameters, and use experimental and field data to validate the model's effectiveness in simulating water hammer pressure transients. First, accounting for fluid compressibility effects, we derive the governing equations for water hammer phenomena, including both the motion equation and continuity equation. Second, based on the mass balance principle and considering perforation friction and fluid filtration effects, we establish boundary constraints for the bottomhole fracture. Third, we develop a new water hammer pressure calculation model by integrating fluid compressibility, tubing friction, perforation friction, and fluid filtration effects within the hydraulic fracture, thereby enabling fracture geometry evaluation. Finally, using a field case study, we conduct a comparative analysis between the model's fracture geometry results and those from microseismic monitoring. The average relative error of the fitted field-monitored water hammer pressure is approximately 0.5%. Compared with the microseismic monitoring results, the average relative errors in fracture half-length and height are 15.4 m and 4.2 m, respectively. Moreover, we highlight the influence of fracture and wellbore-related parameters on water hammer pressure. In this study, we propose a novel water hammer modeling approach capable of fracture geometry evaluation and deepen our fundamental understanding of fracture geometry evaluation using water hammer pressure.

Introduction

In recent years, the rapid development of hydraulic fracturing technology has significantly enhanced the extraction efficiency of oil and gas resources (Scanlan et al. 2018; Wang et al. 2020). The geometry of fractures formed post-fracturing exerts a profound influence on the ultimate recovery rate of reservoirs, making the assessment of hydraulic fracture geometry crucial for optimizing hydrocarbon resource development (Dong et al. 2019; Sadykov et al. 2019; Guo et al. 2021). However, direct observation of subsurface hydraulic fracture geometries after fracturing remains unattainable, necessitating the advancement of cost-effective post-fracturing evaluation techniques.

Fracture geometry assessment methods based on water hammer pressure are gaining widespread attention due to their short testing time and low cost. This method, by analyzing the water hammer phenomenon, can rapidly evaluate the geometry of fractures post-fracturing, enabling developers to effectively assess and optimize fracturing operations. Compared with conventional detection methods, the water hammer pressure-based fracture geometry assessment method not only enhances fracture detection efficiency but also reduces economic costs, meeting the requirements for efficient and low-cost technologies (Clark et al. 2018; Qiu et al. 2022; Hu et al. 2022).

After stopping the pumping of fracturing fluid at each hydraulic fracturing stage, a pressure pulse is formed due to the sudden change in flow rate, propagating back and forth between the wellhead and the bottom of the hole (Clark et al. 2018; Luo et al. 2023). The water hammer pressure analysis method is based on surface oscillatory pressures monitored at the wellhead to analyze the period, amplitude, and decay characteristics of the oscillatory pressures and to quantitatively describe the fracture parameters. Under the condition that the well structure and fluid properties are known, the water hammer signal characteristics are mainly affected by the hydraulic fracture geometry. Previous scholars have carried out a lot of studies on the interpretation of hydraulic fracturing fracture characteristics using water hammer pressure. Streeter and Wylie (1967) and Wylie and Streeter (1993) used the method of characteristics to solve water hammer equations, enhancing solution efficiency and laying the foundation for fracture diagnosis using water hammer phenomena. Holzhausen and Gooch (1985) applied impedance analysis to diagnose geometric parameters of bottomhole fractures. Yew and Ashour (1996) validated the diagnostic effectiveness of impedance analysis for fracture geometric parameters through experimental analysis and field application. Mondal (2010) developed a numerical model of water hammer pressure based on the water hammer equations of motion and continuity equations, which coupled the wellhead boundary and the bottomhole fracture boundary and regarded the hydraulic fracture as a resistive, capacitive, and inertial (RCI) circuit in series and then evaluated the fracture parameters (hereafter referred to as the RCI model). Subsequently, Carey et al. (2015) used the RCI model to match field-acquired water hammer data and evaluate fracture geometric parameters. Hwang et al. (2017) evaluated the influence of varying circuit parameters on water hammer pressure response characteristics. Haustveit et al. (2017) adopted multiple methods to characterize downhole fractures, and the diagnostic results indicate that fracture half-lengths determined by water hammer analysis are similar to those obtained from microseismic analysis. Ma et al. (2019) treated fractures as the well bottom boundary and established a numerical model for water hammer pressure that considers perforation friction and pressure

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leakoff to evaluate fracture-related parameters. Dung et al. (2021) proposed a hydrodynamics-based water hammer response simulator to qualitatively analyze the relationship between water hammer characteristics and fracture characteristics and well productivity. Qiu et al. (2022) used a transient flow model to describe the fluctuation of water hammer pressure and analyzed the effect of fracture on the response characteristics of water hammer signals from multiple perspectives, such as the time, frequency, and quefrency domain. Hu et al. (2022) proposed a new model for calculating the geometry of multiple fractures by considering the stress interference between fractures, thereby evaluating the geometrical characteristics of multiple fractures based on the water hammer signals. These studies have advanced the development and application of water hammer theory. Investigations as detailed above explicitly confirm that water hammer pressure analysis can effectively identify the geometric parameters of hydraulic fractures.

However, most of the above models characterize the bottomhole boundary between the wellbore and fractures as a series-connected circuit composed of resistance, capacitance, and inductance. There is a lack of direct physical correspondence between model parameters (e.g., resistance and capacitance) and fracture geometry. Moreover, the aforementioned models neglect variations in fluid density. During water hammer phenomena, abrupt pressure fluctuations can induce changes in fluid density. Establishing physically meaningful correlation equations to describe the bottomhole fracture boundary and developing a water hammer model that accounts for multiple factors represent highly challenging tasks.

In this study, we propose a novel water hammer modeling approach. First, accounting for fluid compressibility effects, we derive the governing equations for water hammer phenomena, including both the motion equation and continuity equation. Second, based on the mass balance principle and considering perforation friction and fluid filtration effects, boundary constraints for the bottomhole fracture are established. Finally, a new water hammer pressure calculation model is developed by integrating fluid compressibility, tubing friction, perforation friction, and fluid filtration effects within the hydraulic fracture, thereby enabling fracture geometry evaluation. Meanwhile, based on actual field fracturing parameters, we analyze the influencing factors of water hammer pressure curves to guide subsequent fracture geometry evaluation using water hammer pressure data. In contrast to equivalent-circuit-theory-based water hammer modeling approaches, our model defines the wellbore-fracture boundary conditions by rigorously accounting for both perforation friction losses and fracture fluid mass conservation, thereby providing more physically meaningful representations. Moreover, in practical water hammer analysis, the wave speed is often difficult to determine accurately. The proposed model eliminates the need for prior knowledge of the water hammer wave speed while still enabling reliable simulation of pressure transients.

Mathematical Modeling

The following assumptions are adopted for the model in this study: (1) The fluid is compressible; (2) the elastic effects of pipe walls are considered; (3) the frictional resistance along the wellbore is considered; (4) the nonuniformity of flow velocity distribution across the pipe cross section is neglected.

Water Hammer Continuity Equation. Take a pipe section microelement in the wellbore as a control body as shown in **Fig. 1**, where ρ is the fluid density in this control body, v is the fluid flow rate, A is the cross-sectional area of the pipe, dt is the time microelement, ds is the distance microelement in the direction of the pipe, and θ is the angle between the pipe and the horizontal; the inflow mass of this control body, m_1 , can be expressed as (Mondal 2010)

$$m_1 = \rho v A dt. \quad (1)$$

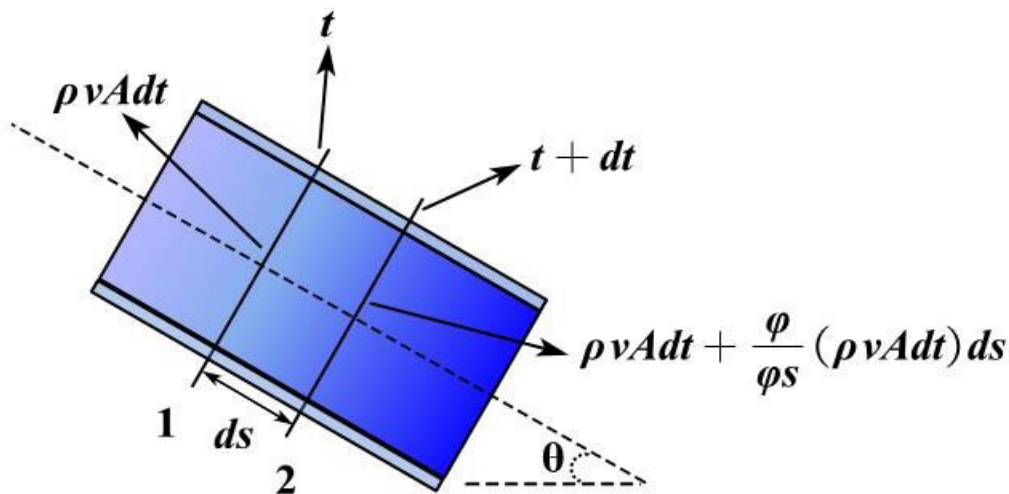


Fig. 1—Microelement analysis of water hammer continuity equation.

The outflow mass m_2 of this control body in time period dt can be expressed as

$$m_2 = \rho v A dt + \frac{\partial}{\partial s} (\rho v A dt) ds. \quad (2)$$

Then, the mass difference between the inflow and outflow of this control body can be obtained as

$$dms = m_1 - m_2 = -\frac{\partial}{\partial s} (\rho v A dt) ds. \quad (3)$$

The change in mass of this control body during the dt time period can be expressed by the following equation:

$$dmt = \frac{\partial}{\partial t} (\rho A ds) dt. \quad (4)$$

According to the theorem of conservation of mass, the mass difference between inflow and outflow should be numerically equal to the value of the change in mass of this control body during the same time period, which leads to the following equation:

$$\frac{v}{\rho} \frac{\partial \rho}{\partial s} + \frac{\partial v}{\partial s} + \frac{v}{A} \frac{\partial A}{\partial s} + \frac{1}{\rho} \frac{\partial \rho}{\partial t} + \frac{1}{A} \frac{\partial A}{\partial t} = 0. \quad (5)$$

Considering that the elasticity of the pipe and the compressibility of the fluid will affect the water hammer, the fluid density ρ and the cross-sectional area A of the pipe are considered as a function of the length of the pipe section s and the time t . The above equation can be deformed as follows:

$$\frac{1}{\rho} \frac{d\rho}{dt} + \frac{1}{A} \frac{dA}{dt} + \frac{\partial v}{\partial s} = 0. \quad (6)$$

In the process of conducting the water hammer study, considering the compressibility of the fluid in the pipeline, the fluid density in the pipeline will be affected due to the dramatic change in pressure, and the fluid density will be considered as a variable for derivation. From the relationship between the fluid density and the fluid elasticity coefficient K , it is obtained that

$$\frac{1}{\rho} \frac{d\rho}{dp} = \frac{1}{K}. \quad (7)$$

The above equation can be transformed into

$$\frac{1}{\rho} \frac{d\rho}{dt} = \frac{1}{K} \frac{dp}{dt}. \quad (8)$$

In engineering, the water head H is commonly used to represent the magnitude of pressure. The pressure can be rewritten as a water head H using the formula $p = \rho g(H - Z)$, where the relationship between density and water head H is derived from the formula involving density:

$$\frac{d\rho}{dt} = \frac{\rho}{\frac{K}{\rho g} - (H - Z)} \frac{d(H - Z)}{dt}. \quad (9)$$

The elasticity coefficient of the pipe is expressed by the symbol E , and its value is the ratio of stress to strain, where the change in stress is $d\sigma$ and the corresponding strain is dD/D . The following equation is obtained:

$$E = \frac{d\sigma}{\frac{dD}{D}} = \frac{D}{dD} d\sigma. \quad (10)$$

The pipe wall thickness is e . The following relationship exists between the pipe wall thickness and the stresses:

$$e = \frac{pD}{2\sigma}. \quad (11)$$

From this, we can obtain

$$d\sigma = \frac{D}{2e} dp. \quad (12)$$

Bringing Eq. 12 into Eq. 10,

$$E = \frac{D^2}{2edD} dp. \quad (13)$$

Owing to:

$$\frac{dA}{A} = \frac{dD^2}{D^2} = \frac{2dD}{D}. \quad (14)$$

Combining Eq. 13 with Eq. 14:

$$\frac{1}{A} \frac{dA}{dp} = \frac{D}{Ee} = \frac{2\sqrt{A}}{\sqrt{\pi Ee}}. \quad (15)$$

Combining Eq. 8 with Eq. 15, Eq. 15 can be rewritten as

$$\frac{1}{A} \frac{dA}{dt} = \frac{2\sqrt{A}}{\sqrt{\pi Ee}} \frac{dp}{dt} = \frac{2K\sqrt{A}}{\sqrt{\pi Ee}} \frac{1}{\rho} \frac{d\rho}{dt}. \quad (16)$$

Bringing Eq. 16 into Eq. 6:

$$\left(1 + \frac{2K\sqrt{A}}{\sqrt{\pi Ee}}\right) \frac{1}{\rho} \frac{d\rho}{dt} + \frac{\partial v}{\partial s} = 0. \quad (17)$$

Bringing Eq. 9 into Eq. 17:

$$\frac{d(H-Z)}{dt} + \frac{\frac{K}{\rho g} - (H-Z)}{1 + \frac{2K\sqrt{A}}{\sqrt{\pi Ee}}} \frac{\partial v}{\partial s} = 0. \quad (18)$$

The differential of the water head with respect to time is given by the equation

$$\frac{d(H-Z)}{dt} = \frac{\partial H}{\partial s} v + \frac{\partial H}{\partial t} - \frac{\partial Z}{\partial s} v - \frac{\partial Z}{\partial t}. \quad (19)$$

Since $\frac{\partial Z}{\partial t} = 0$, $\frac{\partial Z}{\partial s} = -\sin \theta$, brought into Eq. 19, Eq. 19 can be rewritten as

$$\frac{d(H-Z)}{dt} = \frac{\partial H}{\partial s} v + \frac{\partial H}{\partial t} + v \sin \theta. \quad (20)$$

Bringing Eq. 20 into Eq. 18, the water hammer continuity equation considering fluid compressibility is obtained as

$$\frac{\partial H}{\partial s} v + \frac{\partial H}{\partial t} + v \sin \theta + \frac{\frac{K}{\rho g} - (H-Z)}{1 + \frac{2K\sqrt{A}}{\sqrt{\pi Ee}}} \frac{\partial v}{\partial s} = 0. \quad (21)$$

To simplify the form of the equation, let

$$C_M = \frac{\frac{1}{K} + \frac{2\sqrt{A}}{\sqrt{\pi Ee}}}{\frac{1}{\rho g} - \frac{(H-Z)}{K}}. \quad (22)$$

The water hammer continuity equation considering fluid compressibility can be rewritten as

$$\frac{\partial H}{\partial t} + \frac{\partial H}{\partial s} v + v \sin \theta + \frac{1}{C_M} \frac{\partial v}{\partial s} = 0. \quad (23)$$

Water Hammer Equation of Motion. Take a microelement in the pipe section as shown in **Fig. 2**, the cross-sectional area of the microelement is dA , the length is ds , the density of the fluid is ρ , the angle between the pipe axis and the horizontal line is θ , analyze the force on the microelement section, the gravitational component of the fluid mass in the s direction of the microelement section, dGs , is (Mondal 2010)

$$dGs = \rho g dA ds \sin \theta = -\rho g dA ds \frac{\partial Z}{\partial s}. \quad (24)$$

The pressure difference between the two ends of the microelement segment is given by the equation:

$$p dA - \left(p + \frac{\partial p}{\partial s} ds\right) dA = -\frac{\partial p}{\partial s} ds dA. \quad (25)$$

The diameter of the microelement segment is d , and the average stress acting on the cylindrical surface around the microelement segment is τ , then the resistance on the microelement ds segment is $-d\tau\pi d$. u is the flow velocity in the s direction, and for unsteady flow, $u = u(s, t)$, then the acceleration of the microelement segment can be expressed by the following equation:

$$a = \frac{du}{dt} = \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial s}. \quad (26)$$

According to Newton's second law, $F = ma$, and combining Eqs. 24 through 26, the following equation can be obtained:

$$\frac{\partial Z}{\partial s} + \frac{1}{\rho g} \frac{\partial p}{\partial s} + \frac{1}{\rho g} \frac{1}{d} 4\tau + \frac{1}{g} \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial s} \right) = 0. \quad (27)$$

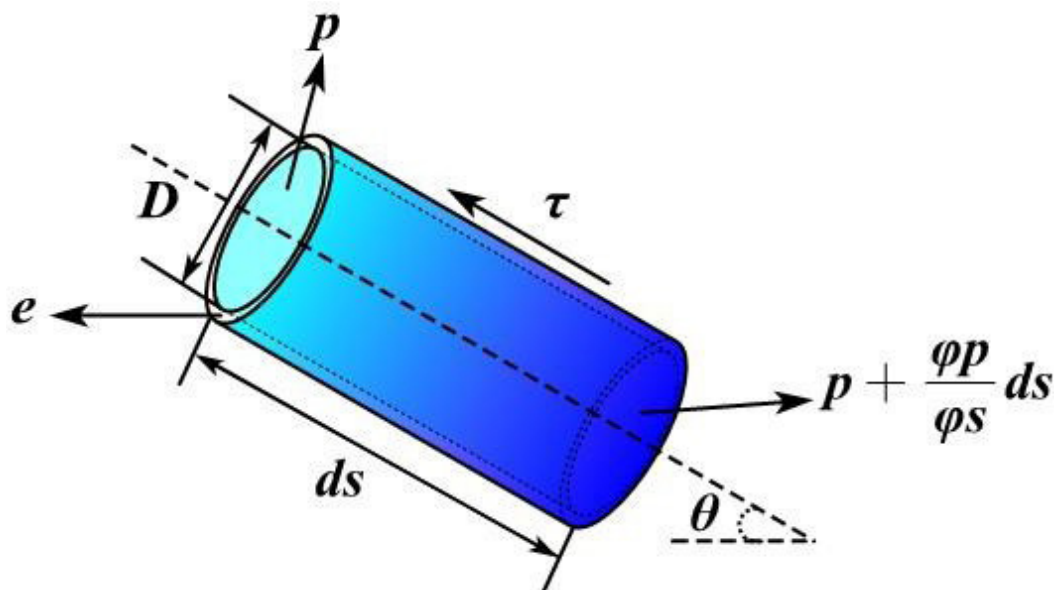


Fig. 2—Microelement analysis of the equations of motion.

The above equation is the differential equation of motion for unsteady flow.

For the total flow, if it is a slow-varying flow, ignoring the inhomogeneity of the flow velocity distribution on the pipe cross section and expressing it in terms of the average flow velocity of the section, the equation of motion for the total flow of the univariate unsteady flow is obtained as follows:

$$\frac{\partial Z}{\partial s} + \frac{1}{\rho g} \frac{\partial p}{\partial s} + \frac{1}{\rho g} \frac{1}{D} 4\tau_0 + \frac{1}{g} \left(\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial s} \right) = 0, \quad (28)$$

$$\tau_0 = \frac{\rho f v |v|}{8}. \quad (29)$$

Bringing Eq. 29 into Eq. 28, we get

$$\frac{\partial Z}{\partial s} + \frac{1}{\rho g} \frac{\partial p}{\partial s} + \frac{f v |v|}{2gD} + \frac{1}{g} \left(\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial s} \right) = 0. \quad (30)$$

Since

$$H = Z + \frac{p}{\rho g}, \quad (31)$$

we can obtain the following equation:

$$\frac{\partial H}{\partial s} = \frac{\partial Z}{\partial s} + \frac{1}{\rho g} \frac{\partial p}{\partial s} = \frac{(H - Z)}{\rho} \frac{\partial \rho}{\partial s}. \quad (32)$$

Bringing Eq. 32 into Eq. 30 yields the equation of motion for water hammer considering fluid compressibility:

$$g \frac{\partial H}{\partial s} + \frac{(H - Z) g}{\rho} \frac{\partial \rho}{\partial s} + g \frac{f v |v|}{2gD} + \frac{\partial v}{\partial t} + v \frac{\partial v}{\partial s} = 0. \quad (33)$$

System of Water Hammer Equations. In this study, we consider the effect of fluid compressibility, which results in a system of equations that includes three variable functions: $H(s, t)$, $v(s, t)$, and $\rho(s, t)$. However, the equation of motion and the continuity equation provide only two equations, meaning that the system is not closed. To address this, we introduce the fluid elasticity equation (Eq. 9) into the water hammer equations, allowing the system to be closed. Consequently, the complete system of water hammer equations, which takes fluid compressibility into account, is as follows:

$$\begin{cases} g \frac{\partial H}{\partial s} + \frac{(H-Z)g}{\rho} \frac{\partial \rho}{\partial s} + g \frac{f v |v|}{2gD} + \frac{\partial v}{\partial t} + v \frac{\partial v}{\partial s} = 0 \\ \frac{\partial H}{\partial t} + \frac{\partial H}{\partial s} v + v \sin \theta + \frac{1}{C_M} \frac{\partial v}{\partial s} = 0 \\ \frac{d\rho}{dt} = \frac{\rho}{\frac{K}{\rho g} - (H-Z)} \frac{d(H-Z)}{dt} \end{cases} \quad (34)$$

Solution of the Equation System

For this study, we utilize the water hammer equations, which involve numerous unknown parameters. The finite difference method is adopted as it effectively solves these multivariable partial differential equations. To effectively simulate water hammer pressure under hydraulic fracturing conditions, a wellbore-fracture boundary condition is established based on mass balance principles, accounting for perforation friction and fluid leakoff effects in fractures. This enables the evaluation of fracture geometry parameters through history matching of monitored water hammer pressure.

Difference Equation. The system of water hammer equations derived in this study has more unknowns, and the finite difference method is used to solve the system of equations (Shen and Li 2025). That is, the solution region of the required partial differential equation is first subdivided into a grid consisting of a finite number of gridpoints, and then a finite difference formula is used to replace the derivatives of each gridpoint, and finally, the approximation is carried out to solve the equation, and this process can be regarded as the process of replacing the solution of the partial differential equation with an interpolating polynomial and its differentiation. The finite difference method effectively solves multivariate partial differential equations, although it requires increased computation time.

Y represents the water head (H), flow velocity (v), and density (ρ), and Y_i^j represents the value as a function of section i of layer j at time j . The partial differentiation of Y with respect to time can be expressed as follows:

$$\frac{\partial Y}{\partial t} \approx \frac{Y_i^j - \left[\alpha Y_i^{j-1} + \frac{1-\alpha}{2} (Y_{i+1}^{j-1} + Y_{i-1}^{j-1}) \right]}{\Delta t} \quad (35)$$

The partial differentiation of the distance is expressed as an intermediate difference quotient, and the difference format is as follows:

$$\frac{\partial Y}{\partial s} \approx \frac{Y_{i+1}^{j-1} - Y_{i-1}^{j-1}}{2\Delta s} \quad (36)$$

The three partial differential equations in the system of equations (Eq. 34) are transformed in difference format to obtain the expressions for the three unknowns of water head H , flow velocity v , and fluid density ρ as

$$H_i^j = C_{A1} - v_i^{j-1} \left(\frac{C_{A2}}{2} + \Delta t \sin \theta \right) - \frac{C_{B2}}{2C_M}, \quad (37)$$

$$v_i^j = C_{B1} - \frac{v_i^{j-1}}{2} C_{B2} - \frac{g}{2} C_{A2} - \frac{(H_i^{j-1} - Z_i)g}{2\rho_i^{j-1}} C_{E2} - g R v_i^{j-1} |v_i^{j-1}|, \quad (38)$$

$$\rho_i^j = C_{E1} - \frac{v_i^{j-1}}{2} C_{E2} + \frac{\rho_i^{j-1}}{C_N} \left[H_i^j - C_{A1} + v_i^{j-1} \left(\frac{C_{A2}}{2} + \Delta t \sin \theta \right) \right], \quad (39)$$

where

$$C_{A1} = \alpha H_i^{j-1} + \frac{1-\alpha}{2} (H_{i+1}^{j-1} + H_{i-1}^{j-1}),$$

$$C_{A2} = (H_{i+1}^{j-1} - H_{i-1}^{j-1}) \frac{\Delta t}{\Delta s},$$

$$C_{B1} = \alpha v_i^{j-1} + \frac{1-\alpha}{2} (v_{i+1}^{j-1} + v_{i-1}^{j-1}),$$

$$C_{B2} = (v_{i+1}^{j-1} - v_{i-1}^{j-1}) \frac{\Delta t}{\Delta s},$$

$$C_{E1} = \alpha \rho_i^{j-1} + \frac{1-\alpha}{2} (\rho_{i+1}^{j-1} + \rho_{i-1}^{j-1}),$$

$$C_{E2} = (\rho_{i+1}^{j-1} - \rho_{i-1}^{j-1}) \frac{\Delta t}{\Delta s},$$

$$C_N = \frac{K}{\rho_i^{j-1} g} - (H_i^{j-1} - Z_i),$$

$$C_M = \left(1 + \frac{2K\sqrt{A}}{\sqrt{\pi} E e} \right) / C_N,$$

$$R = \frac{f\Delta t}{2gD}.$$

Initial Conditions. When the pump begins to be stopped, the fluid flow within the wellbore is considered to be under steady-state tubular flow conditions, and the flow rate is uniform throughout the wellbore. Regarding the density distribution within the wellbore, it is assumed that the density at every point within the wellbore at the initial moment is ρ_0 . Then, the pressure distribution along the wellbore at the initial moment is obtained by using $p = \rho g(H - Z)$, and then the pressure distribution is substituted into Eq. 7 to obtain the density distribution along the wellbore. The circulation is repeated multiple times until the density and pressure distributions along the wellbore are stabilized, which is considered to be the density distribution at the initial moment. Additionally, at the initial moment, the water head at the wellhead is known, and the water head distribution at each point within the wellbore can be calculated using the Darcy-Weisbach formula (Brown 2002), which yields the following equation:

$$Q_i^0 = Q_0, \quad (40)$$

$$H_1^0 = H_0, \quad (41)$$

$$\Delta H = f \frac{\Delta L Q^2}{2gDA^2}, \quad (42)$$

The wellbore friction coefficient can be calculated by the following formula (Chen 1979):

$$\frac{1}{\sqrt{f}} = -2 \log \left[\frac{\varepsilon}{3.7065D} - \frac{5.0452}{Re} \cdot \log \left(\frac{1}{2.8257} \left(\frac{\varepsilon}{D} \right)^{1.1098} + \frac{5.8506}{Re^{0.8981}} \right) \right]. \quad (43)$$

Boundary Conditions. Boundary conditions include two categories. One category is the flow boundary at the wellhead, and the other is the fracture boundary at the bottom of the well.

Wellhead Boundary. The flow rate at the wellhead changes with time and can be expressed by the following equation:

$$Q_1^t = Q(t). \quad (44)$$

Based on the wellhead flow boundary conditions, the numerical solution format for Node Cell 1 at the wellhead can be obtained in the following format:

$$Q_1^{t+\Delta t} = Q(t + \Delta t). \quad (45)$$

Wellbore Bottom Boundary. A schematic of the fracture boundary is shown in Fig. 3. For a multicluster fracturing section, it is usually assumed that a fracturing cluster may open multiple perforations and form one main fracture, or no fracture may open. Within a fracturing section, the number of effectively opened main fracture clusters is denoted as N_f , and the number of effectively opened perforations is denoted as N_p , assuming that the opened main fracture is a planar slit with an average fracture half-length of l_f and an average fracture height of h_f . The planar fracture assumption is primarily applicable to deep shale gas reservoirs with high horizontal principal stresses. However, it is less suitable for describing the complex fracture geometries typically observed in shallow shale gas reservoirs. Combining the perforation friction resistance (Massarar et al. 2007; He et al. 2024) and the relationship of fluid mass conservation within the fracture, the following boundary conditions for the fracture in the bottom of a well can be obtained as follows:

$$p_w - p_f = \frac{8\rho Q^2}{\pi^2 N_p^2 d_p^4 C_D^2}, \quad (46)$$

$$2N_f h_f \frac{dw_f}{dt} = Q - Q_{leak}. \quad (47)$$

The hydraulic fracture opening w_f can be related to the fluid pressure in the fracture and the fracture height by the following equation (Paige et al. 1995), with the following representation:

$$w_f = \frac{2(p_f - \sigma_{min}) h_f (1 - \nu)}{\pi G \alpha_s}. \quad (48)$$

The fluid filtration loss within the fracture can be calculated by the following equation (Howard and Fast 1957):

$$Q_{leak} = \frac{2N_f h_f C_L}{\sqrt{f_f}}. \quad (49)$$

For the well bottom boundary, the water hammer equation set contains three equations, and two more equations (Eqs. 46 and 47) are added to the well bottom fracture constraint equations; however, the unknowns in the five equations are the water head H , the flow velocity v , the fluid density ρ , and the pressure p_f inside the fracture, which results in a contradiction of more equations than unknowns. For this reason, the continuity equations and the equations of motion are synthesized into one partial differential equation, which is then transformed into a difference equation associating other equations for solution.

Multiplying the equation of motion (Eq. 33) by v_0/g and adding it to the equation of continuity (Eq. 23) yields

$$\frac{\partial H}{\partial t} + (v + v_0) \frac{\partial H}{\partial s} + \frac{v_0}{g} \frac{\partial v}{\partial t} + \left(\frac{1}{C_M} + \frac{v_0 v}{g} \right) \frac{\partial v}{\partial s} + \frac{(H - Z) v_0}{\rho} \frac{\partial \rho}{\partial s} + v \sin \theta + v_0 \frac{f|v|}{2gD} = 0. \quad (50)$$

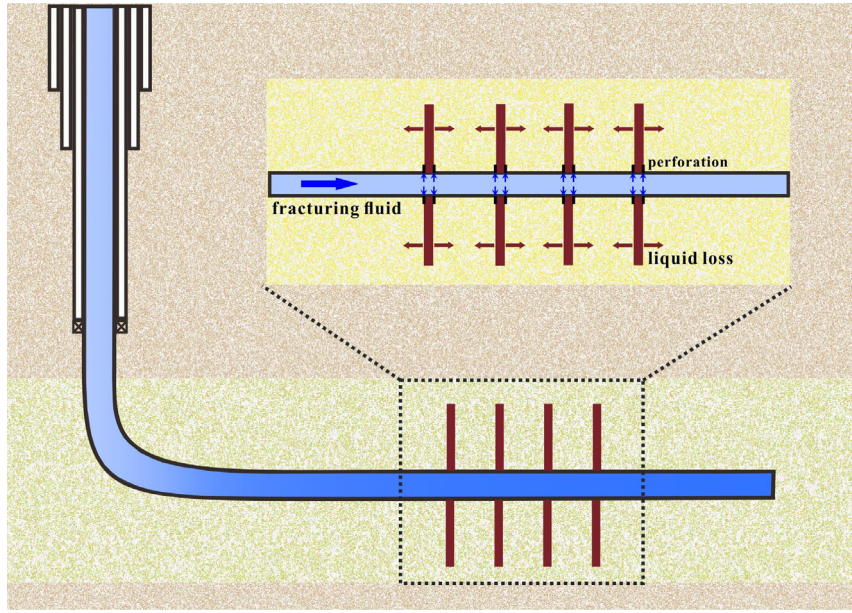


Fig. 3—Schematic of horizontal well segmented multicluster fracturing.

By differencing Eqs. 34 and 50, we obtain

$$\begin{aligned} & \frac{H_i^j - H_{i-1}^{j-1}}{\Delta t} + (v_i^j + v_0) \frac{(H_i^j - H_{i-1}^j)}{\Delta s} + \frac{v_0}{g} \frac{(v_i^j - v_{i-1}^{j-1})}{\Delta t} + \left(\frac{1}{C_{M1}} + \frac{v_0}{g} v_i^j \right) \frac{(v_i^j - v_{i-1}^j)}{\Delta s} \\ & + \frac{(H_i^j - Z_i) v_0 (\rho_i^j - \rho_{i-1}^j)}{\rho_i^j \Delta s} + v_i^j \sin \theta + v_0 \frac{f v_i^j |v_i^j|}{2gD} = 0 \end{aligned} \quad (51)$$

$$C_{M1} = \frac{\frac{1}{K} + \frac{2\sqrt{A}}{\sqrt{\pi} E e}}{\frac{1}{\rho_i^j g} - \frac{(H_i^j - Z_i)}{K}}, \quad (52)$$

$$\rho_i^j - \rho_{i-1}^{j-1} + v_i^{j-1} (\rho_{i-1}^{j-1} - \rho_{i-2}^{j-1}) \frac{\Delta t}{\Delta s} - \frac{\rho_{i-1}^{j-1}}{C_N} \left[H_i^j - H_{i-1}^{j-1} + v_i^{j-1} \left((H_{i-1}^{j-1} - H_{i-2}^{j-1}) \frac{\Delta t}{\Delta s} + \Delta t \sin \theta \right) \right] = 0. \quad (53)$$

The water head, flow rate, and density of the wellbore nodal cell n can be derived by solving Eqs. 51 and 53, and combining this with the wellbore-fracture boundary conditions (Eqs. 46 and 47) results in the following format:

$$\begin{cases} \frac{H_n^j - H_{n-1}^{j-1}}{\Delta t} + (v_n^j + v_0) \frac{(H_n^j - H_{n-1}^j)}{\Delta s} + \frac{v_0}{g} \frac{(v_n^j - v_{n-1}^{j-1})}{\Delta t} + \left(\frac{1}{C_{M1}} + \frac{v_0}{g} v_n^j \right) \frac{(v_n^j - v_{n-1}^j)}{\Delta s} \\ + \frac{(H_n^j - Z_n) v_0 (\rho_n^j - \rho_{n-1}^j)}{\rho_n^j \Delta s} + v_n^j \sin \theta + v_0 \frac{f v_n^j |v_n^j|}{2gD} = 0 \\ \rho_n^j - \rho_{n-1}^{j-1} + v_n^{j-1} (\rho_{n-1}^{j-1} - \rho_{n-2}^{j-1}) \frac{\Delta t}{\Delta s} - \frac{\rho_{n-1}^{j-1}}{C_N} \left[H_n^j - H_{n-1}^{j-1} + v_n^{j-1} \left((H_{n-1}^{j-1} - H_{n-2}^{j-1}) \frac{\Delta t}{\Delta s} + \Delta t \sin \theta \right) \right] = 0 \\ p_n^j - p_f^j = \frac{8\rho_n^j}{\pi^2 N_p^2 d_p^4 C_D^2} (Q_n^j)^2 \\ 2N_f h_{fj} (w_f^j - w_f^{j-1}) = (Q_n^j - Q_{leak}) \Delta t \end{cases} \quad (54)$$

The water hammer pressure calculation process is shown in Fig. 4.

Results and Discussion

To validate the water hammer pressure model developed in this study, we verify its effectiveness in simulating pressure transients using both experimental data from fracture-equipped systems and field-monitored water hammer data. In addition, using the water hammer model developed in this study, we assess fracture geometry by fitting monitored water hammer pressure data from multistage fracturing operations in a horizontal well in the Sichuan Basin.

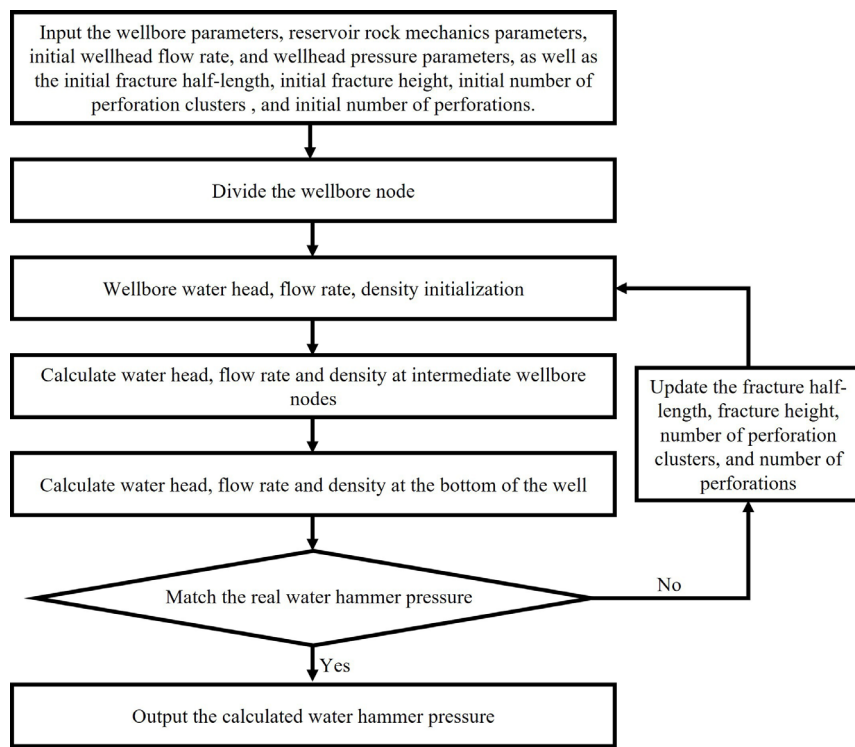


Fig. 4—Flow chart for water hammer pressure calculation.

Model Validation. Based on the experimental water hammer pressure data considering fractures from Hu et al. (2022), this study calculates the water hammer pressure using our proposed model and compares the results with experimental data to validate the model's effectiveness. The experimental setup simulates fracture effects by connecting a pressure vessel through a branch pipe. The experimental parameters are as follows: The horizontal pipe length is 157 m, the internal diameter of the pipe is 25 mm, the pipe friction coefficient is 0.0376, the initial flow rate within the pipe is 0.0324 m³/min, the pump shutdown duration is 5.9 seconds, and the pressure vessel pressure is 0.052 MPa. As shown in Fig. 5, the model predictions agree well with experimental results, demonstrating a maximum pressure deviation of only 2.42%. This close agreement validates the model's effectiveness in simulating water hammer pressure evolution.

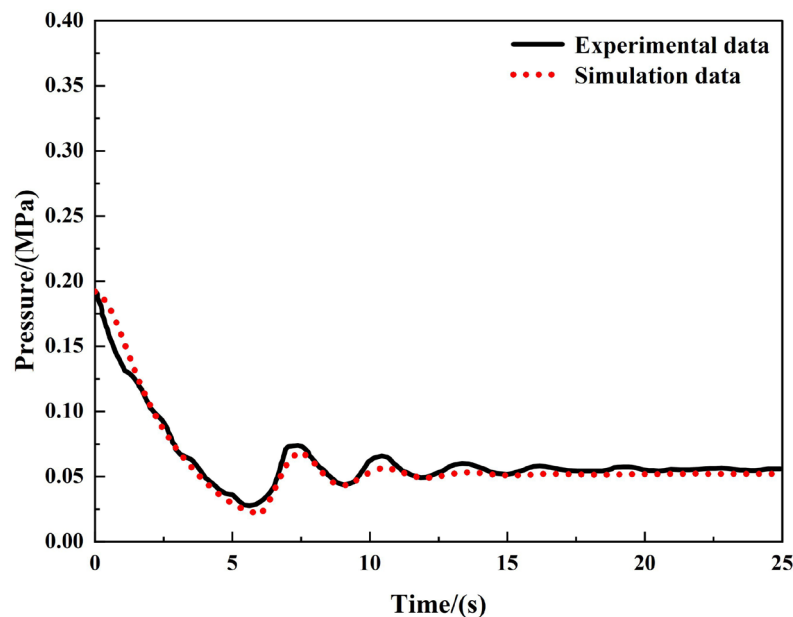


Fig. 5—Comparison of model-calculated data with experimental data.

Using fracturing parameters from a shale gas horizontal well (Well A) in the Sichuan Basin (He et al. 2024), we calculate the water hammer pressure curve with our model and compare it with field-measured data from Well A, the basic parameters of which are listed in Table 1.

Parameters	Values	Parameters	Values
Length of the vertical well section (m)	3180	Tube Young's modulus (GPa)	206
Length of the horizontal well section (m)	2120	Tube Poisson's ratio	0.3
Inner diameter of wellbore (mm)	118.6	Rock Young's modulus (GPa)	42
Wall thickness (mm)	10.54	Rock Poisson's ratio	0.22
Fracture half-length (this study) (m)	160	Fracture half-length (He et al.) (m)	180
Fracture height (this study) (m)	19	Fracture height (He et al.) (m)	20

Table 1—Base parameters of Well A.

Fig. 6 reveals that the water hammer pressure curve exhibits periodic oscillatory attenuation. The early stage shows significant pressure decay, while the later-stage attenuation amplitude decreases under the combined effects of wellbore-perforation friction, fracture opening/closure, and fluid loss, eventually stabilizing. The water hammer pressure curve calculated by the model developed in this study exhibits a mean relative error of only 0.23% compared with the monitored water hammer pressure curve, indicating a high degree of consistency. Based on the actual fracturing parameters, the water hammer model established in this study effectively simulates water hammer pressure variations. Both experimental data and field monitoring data demonstrate that the model developed in this study can accurately simulate water hammer pressure changes.

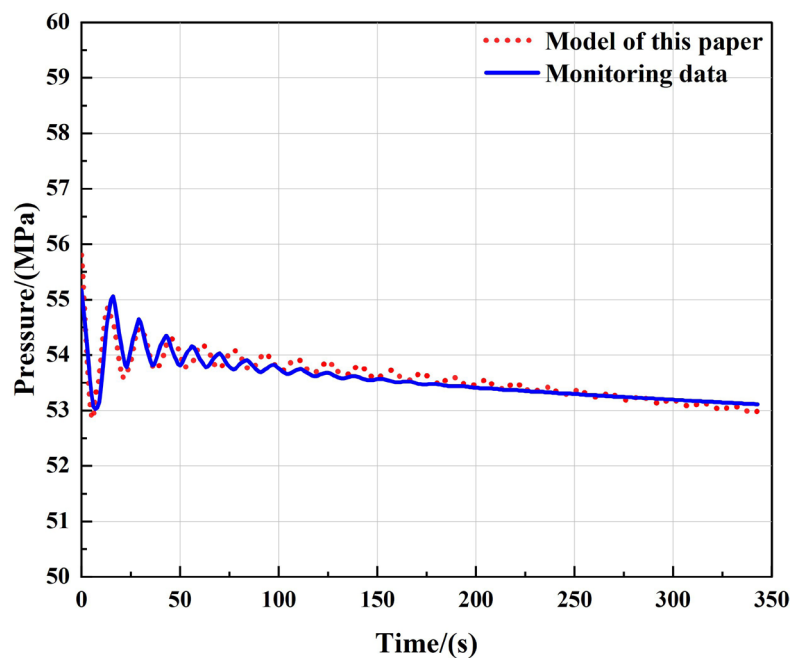


Fig. 6—Comparison between calculated and actual results.

Sensitivity Analysis. Analyzing the influence of fracture and wellbore parameters on water hammer pressure is crucial for guiding the use of water hammer pressure to interpret fracture parameters. By examining the influencing factors in the early stages and assessing the shape of water hammer pressure curves under varying parameters, one can effectively estimate the approximate range of fracture parameters, thereby significantly improving the efficiency of fracture parameter calculations. Therefore, based on the basic data of Well B's fracturing section, we analyze the impact of different fracture and wellbore parameters on water hammer pressure. The fundamental parameters are presented in **Table 2**.

Parameters	Values	Parameters	Values
Length of the vertical well section (m)	2926	Fracture half-length (m)	157.4
Length of the horizontal well section (m)	1597	Fracture height (m)	23.7
Rock Young's modulus (GPa)	42	Number of open clusters	5
Rock Poisson's ratio	0.23	Number of open perforations	26

Table 2—Base parameters used in the simulations.

Effects of Fracture Parameters. Effect of Fracture Half-Length. Based on the basic data of Well B, to analyze the influence of fracture half-length on the water hammer pressure curve, fracture half-lengths of 80 m, 140 m, 200 m, and 260 m were designed and substituted into the model of this study to calculate the corresponding water hammer pressure curves, as shown in Fig. 7. From Fig. 7, it is observed that the fracture half-length has a significant impact on the decay rate of the water hammer pressure curve; as the fracture half-length increases, the curve decays more rapidly. The increase in fracture half-length not only enlarges the fluid loss area but also extends the wave propagation path, resulting in greater frictional resistance and ultimately accelerating energy dissipation.

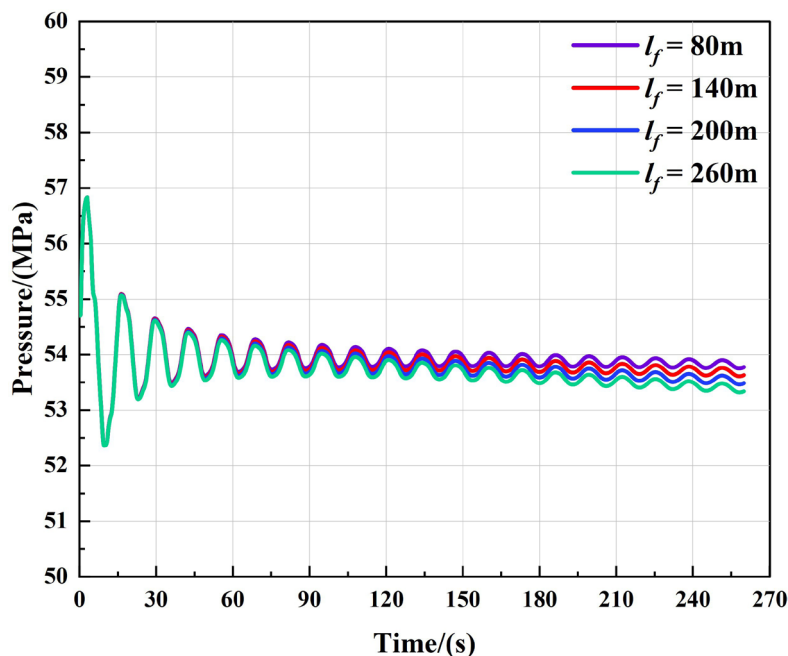


Fig. 7—Water hammer pressure curves for different fracture half-lengths.

Effect of Fracture Height. Taking fracture heights of 10 m, 20 m, 30 m, and 40 m, we calculate the water hammer pressure curves under different fracture heights and analyze their influence on the water hammer pressure curves. Fig. 8 presents the calculation results of water hammer pressure under varying fracture heights. As the fracture height increases, the decay rate of water hammer pressure slows down. The increase in fracture height enhances the fracture's elastic storage capacity, reducing energy loss during fluid injection and release. Moreover, greater fracture height increases the fluid volume within the fracture, strengthening the fluid's propping effect. This results in more uniform pressure distribution within the fracture and relatively lower resistance to pressure wave propagation.

Effect of Perforation Clusters and Perforations. In actual fracturing operations, the competitive initiation among different perforation clusters results in only partial clusters and perforations being activated. Fig. 9 simulates the water hammer pressure under varying numbers of activated perforation clusters and perforations. Analysis of Fig. 9 reveals that fewer activated perforation clusters and perforations lead to an overall lower pressure level. This phenomenon can be attributed to the fact that, under a constant flow rate, a reduced number of activated perforation clusters and perforations not only increase the perforation friction loss but also simplify and concentrate the fluid flow path. This reduction mitigates flow interference among different clusters and perforations, thereby decreasing the complexity of fluid dynamics and ultimately resulting in a significant reduction in pressure levels.

Effect of Leakoff Coefficient. Fluid leakoff effects in fractures significantly influence water hammer pressure during both fracturing injection and pump shutdown processes. Simulations were conducted using leakoff coefficients of $2 \times 10^{-7} \text{ m/s}^{0.5}$, $2 \times 10^{-6} \text{ m/s}^{0.5}$, and $2 \times 10^{-5} \text{ m/s}^{0.5}$, with the computational results presented in Fig. 10, which demonstrates that the leakoff coefficient significantly affects the water hammer pressure decline rate, with larger coefficients resulting in faster pressure decay. On one hand, increased leakoff coefficient enhances fracturing fluid loss to the formation, reducing effective fluid volume in fractures and diminishing the system's energy storage capacity. On the other hand, higher leakoff coefficients facilitate faster pressure dissipation into the formation. Under the combined effects of mass loss and energy dissipation mechanisms, elevated leakoff coefficients accelerate water hammer pressure decline.

Effect of Wellbore Parameters. Effect of Horizontal Wellbore Length. As the fracturing operation advances and the fracturing section progresses, the actual horizontal wellbore length corresponding to the fracturing pump shutdown gradually decreases, necessitating an analysis of the impact of horizontal wellbore length on the water hammer pressure signal. Here, simulations are conducted with horizontal wellbore lengths of 1 km, 2 km, and 3 km, respectively, and the results are displayed in Fig. 11. From Fig. 11, it is observed that the length of the horizontal wellbore affects the water hammer pressure cycle. As the wellbore length increases, the water hammer pressure wave requires more time for its round trip, leading to a larger water hammer pressure oscillation cycle. Additionally, with the increase in wellbore length, wellbore friction increases, and the energy dissipation of the pressure during propagation also increases, resulting in accelerated pressure decay.

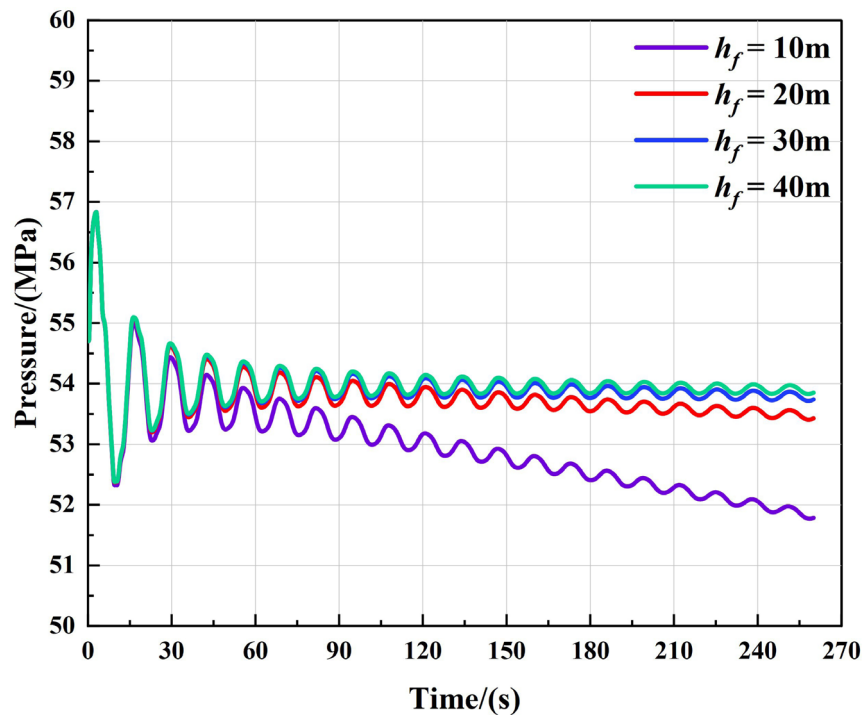


Fig. 8—Water hammer pressure curves at different fracture heights.

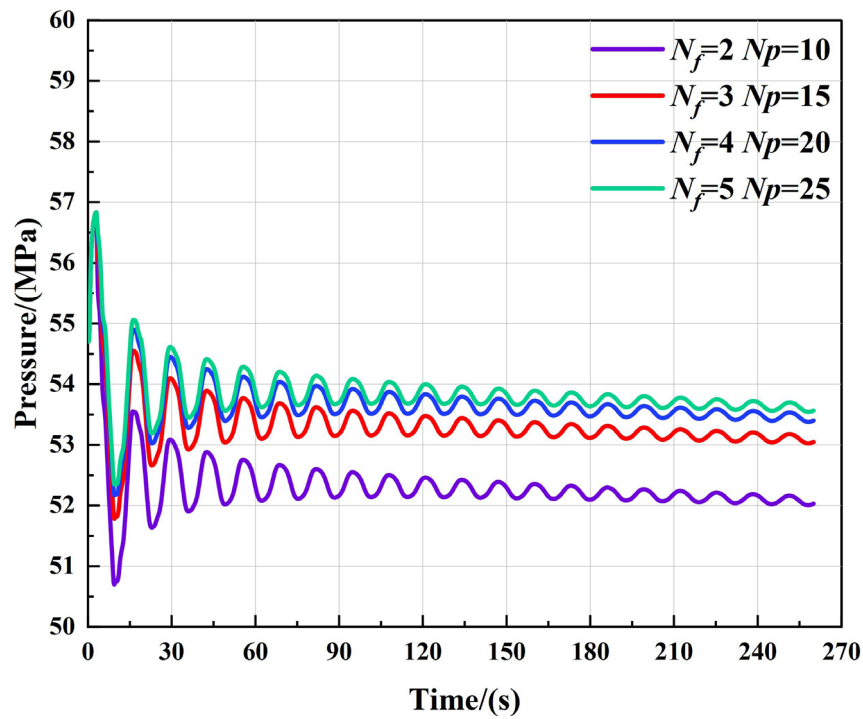


Fig. 9—Water hammer pressure curves for different numbers of perforation clusters and perforations.

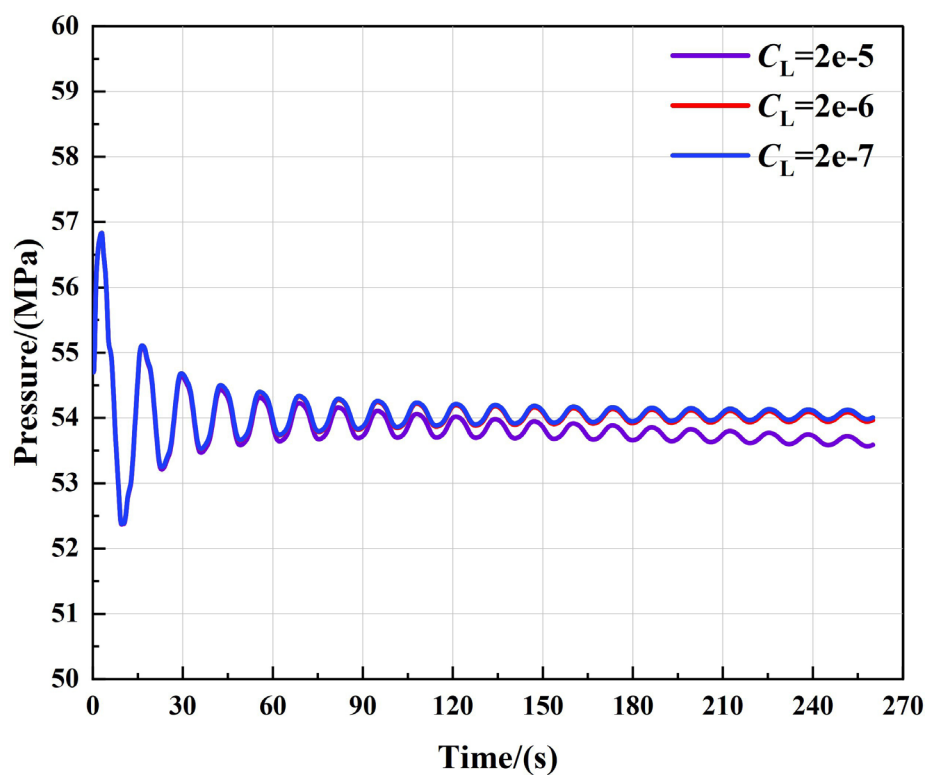


Fig. 10—Water hammer pressure curves with different leakoff coefficients.

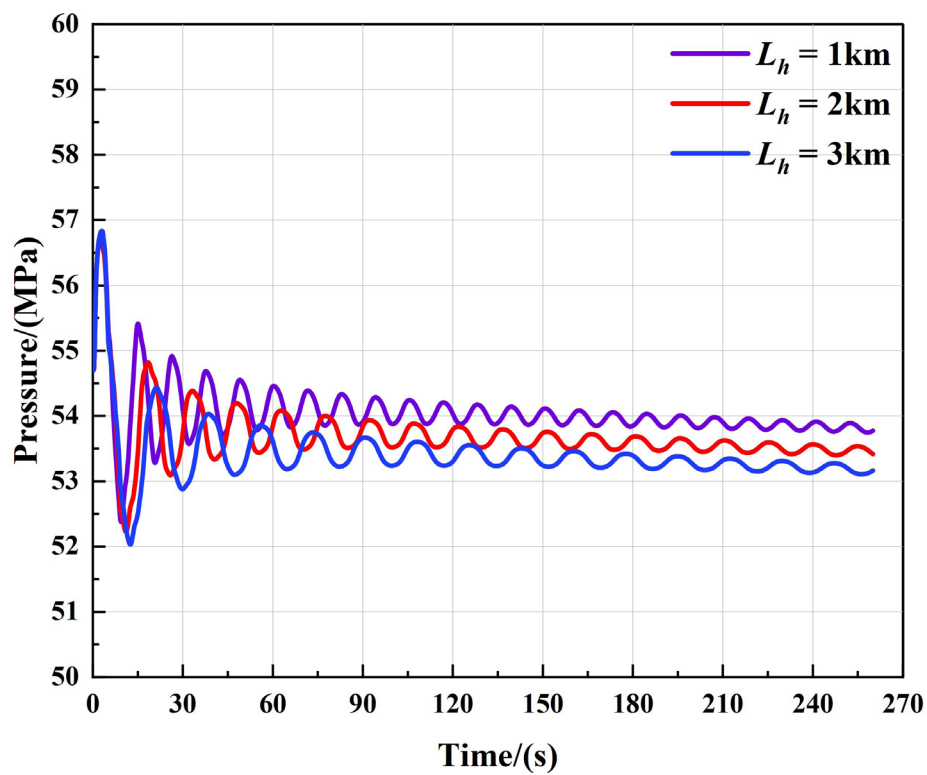


Fig. 11—Water hammer pressure curves at different horizontal wellbore lengths.

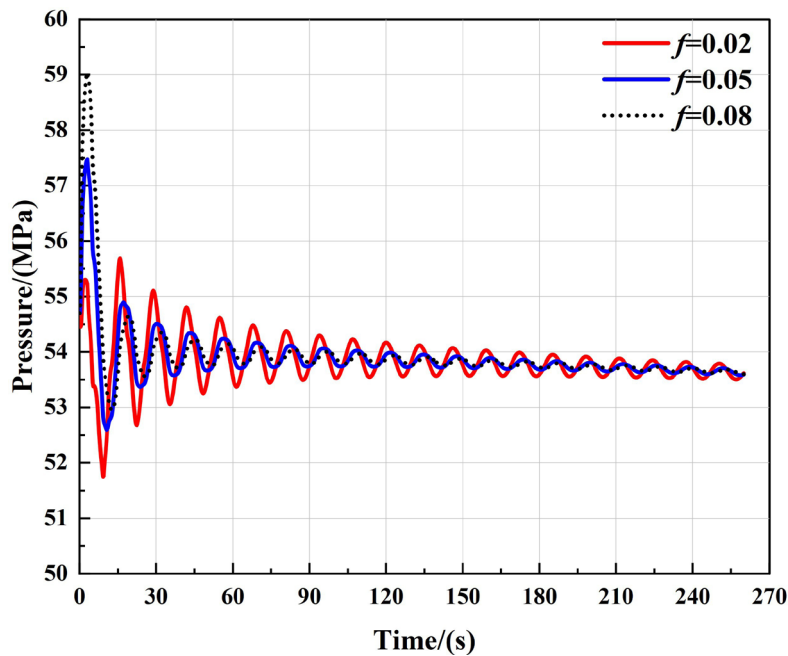


Fig. 12—Water hammer pressure curves for different wellbore along-track friction coefficients.

Effect of Friction Coefficient along the Wellbore. The wellbore friction coefficient is closely related to the resistance of fluid flow along the wellbore and significantly influences the water hammer pressure. To analyze the impact of the wellbore along-travel friction coefficient on water hammer pressure, simulations are conducted with wellbore along-travel friction coefficients of 0.02, 0.05, and 0.08, and the results are depicted in Fig. 12. From Fig. 12, it is evident that along-travel friction affects the decay rate of the water hammer pressure oscillation amplitude. The larger the along-travel friction coefficient, the greater the energy loss during pressure propagation, and the faster the decay of the pressure oscillation amplitude.

Field Case Analysis. Using our model with wellbore and formation parameters from a Sichuan Basin shale gas horizontal well (see Table 3 for complete parameters), we simulated water hammer pressure and matched field-measured data to evaluate fracture geometry.

Parameters	Well B-1	Well B-2	Well B-3	Well B-4
Length of the vertical well section (m)	2926	2926	2926	2926
Length of the horizontal well section (m)	2255	2182	2107	2039
Inner diameter of wellbore (mm)	114.3	114.3	114.3	114.3
Rock Young's modulus (GPa)	42	42	42	42
Rock Poisson's ratio	0.23	0.23	0.23	0.23

Table 3—Base parameters for each fractured section of Well B.

Using stage-specific water hammer pressure data, we iteratively adjusted fracture half-length, fracture height, number of perforation clusters, and number of perforations until achieving satisfactory agreement between simulated and measured pressure responses, thereby determining optimal fracture geometry parameters for each stage.

Analysis of Fig. 13 and Table 4 demonstrates strong agreement between simulated and measured water hammer pressure curves for Well B's four fracturing stages, with mean relative errors of 0.39%, 0.55%, 0.62%, and 0.53%, respectively. A comparative analysis between the fitted fracture parameters and field monitoring results shows mean relative errors of 15.4 m for fracture half-length and 4.2 m for fracture height across all stimulated stages. Additionally, the proposed model enables analysis of activated cluster counts and effective perforation numbers.

Well Section Name	Model of This Study				Microseismic Monitoring	
	l_f (m)	h_f (m)	N_p	N_f	l_f (m)	h_f (m)
Well B-1	152.30	21.40	37	5	166	24
Well B-2	162.80	27.61	34	6	174	31
Well B-3	150.80	22.46	27	5	168	28
Well B-4	178.60	30.40	46	6	159	25

Table 4—Comparison of calculation results for Well B.

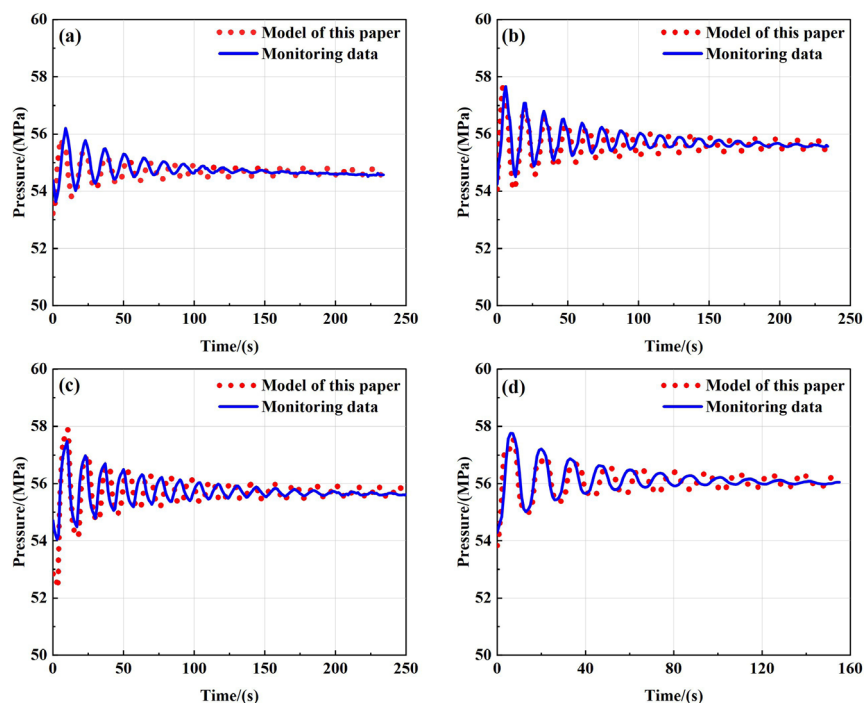


Fig. 13—Comparison of simulated and monitored water hammer curves for Well B (a) for Section 1, (b) for Section 2, (c) for Section 3, and (d) for Section 4.

Discussion. In this study, we propose a novel water hammer modeling approach. First, accounting for fluid compressibility effects, we derive the governing equations for water hammer phenomena, including both the motion equation and continuity equation. Second, based on the mass balance principle and considering perforation friction and fluid filtration effects, boundary constraints for the bottomhole fracture are established. Finally, a new water hammer pressure calculation model is developed by integrating fluid compressibility, tubing friction, perforation friction, and fluid filtration effects within the hydraulic fracture, thereby enabling fracture geometry evaluation. In contrast to equivalent-circuit-based water hammer models, our approach establishes wellbore-fracture boundary conditions through perforation friction and fracture fluid mass conservation, providing clearer physical significance and overcoming the inherent limitation where circuit model parameters lack direct correspondence with fracture geometry. Moreover, in practical water hammer analysis, the wave speed is often difficult to determine accurately. The proposed model eliminates the need for prior knowledge of the water hammer wave speed while still enabling reliable simulation of pressure transients. When evaluating fracture geometry using water hammer pressure, both the RCI model and our proposed model rely on iterative parameter adjustment to achieve satisfactory matching between simulated and measured pressure data. The fracture geometry parameters that produce the best match are considered representative of the actual fracture system. However, the fracture geometry evaluation using water hammer pressure contains inherent uncertainties, as different parameter combinations may produce similar pressure responses during multiparameter estimation. Various hydraulic fracturing monitoring techniques, such as distributed fiber-optic strain monitoring and high-frequency pressure monitoring, can provide prior information on the number of effective perforation clusters and effective perforations, thereby reducing the dimensionality of uncertain parameters. This approach may reduce parameter uncertainty, though further research and data validation are still required. The proposed model in this study provides a theoretical framework for reducing uncertainty in fracture geometry parameters derived from water hammer pressure matching through the aforementioned methodology. Effectively reducing these uncertainties represents a critical research direction for fracture geometry evaluation using water hammer pressure, which will be further investigated in subsequent studies.

Conclusion

Based on mass balance principles and considering both perforation friction and fluid leakoff effects, this study establishes wellbore-fracture boundary conditions to develop a novel water hammer modeling approach. Moreover, we highlighted the influence of fracture- and wellbore-related parameters on water hammer pressure. The following conclusions were obtained:

1. By establishing wellbore-fracture boundary conditions based on perforation friction and fracture fluid mass conservation, while newly incorporating fluid compressibility effects, we develop a comprehensive mathematical model for water hammer pressure evolution that accounts for fluid compressibility, wellbore friction, perforation friction, and fracture fluid leakoff, enabling fracture geometry evaluation.
2. The model-calculated water hammer pressures show excellent agreement with both experimental measurements and field monitoring data, demonstrating reliable simulation of pressure transient evolution.

3. Analysis of influencing factors indicates that longer fracture half-length and higher leakoff coefficient accelerate the water hammer pressure decline, whereas increased fracture height slows it. Fewer active perforation clusters and perforations reduce the overall water hammer pressure. Longer horizontal wellbores prolong the water hammer oscillation period, while a higher friction coefficient accelerates amplitude attenuation.
4. The geometry of the fracture was evaluated by fitting the field-monitored water hammer pressure data, with the average relative error of the water hammer pressure fitting being around 0.5%. Compared with the microseismic monitoring results, the average relative errors in fracture half-length and height were 15.4 m and 4.2 m, respectively.

Nomenclature

A = cross-sectional area, m^2
 C_D = flow rate coefficient
 C_L = filtration loss coefficient, $\text{m/s}^{0.5}$
 d_p = perforation hole diameter, m
 D = inner diameter of wellbore, m
 e = pipe wall thickness, m
 E = Young's modulus, Pa
 f = friction coefficient
 g = acceleration of gravity, m/s^2
 G = shear modulus of elasticity, Pa
 h_f = fracture height, m
 H = water head, m
 H_0 = initial pressure head, m
 H_1 = initial wellhead pressure head, m
 K = bulk modulus, Pa
 l_f = fracture half-length, m
 N_f = number of open clusters
 N_p = number of open perforations
 P = pressure, Pa
 P_f = effective fluid pressure in the fracture, Pa
 P_w = bottomhole pressure, Pa
 Q = wellbore flow rate, m^3/s
 Q_i^0 = initial wellbore node flow rate, m^3/s
 Q_{leak} = filter loss in the fracture, m^3/s
 Q_0 = initial wellhead flow rate, m^3/s
 Re = Reynolds number
 s = pipe length, m
 t = time, seconds
 t_f = filtration reference time, seconds
 v = flow velocity, m/s
 w_f = fracture width, m
 Z = elevation of the reference datum, m
 α = weighting factor, 0.1
 α_s = fracture shape factor
 ε = roughness of wellbore, m
 θ = inclination of the pipe to the horizontal ground, degrees
 ρ = fluid density, kg/m^3
 σ_{min} = minimum horizontal formation stress, Pa
 ν = Poisson's ratio

Acknowledgments

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